GATEHOUSE UX CASE STUDY

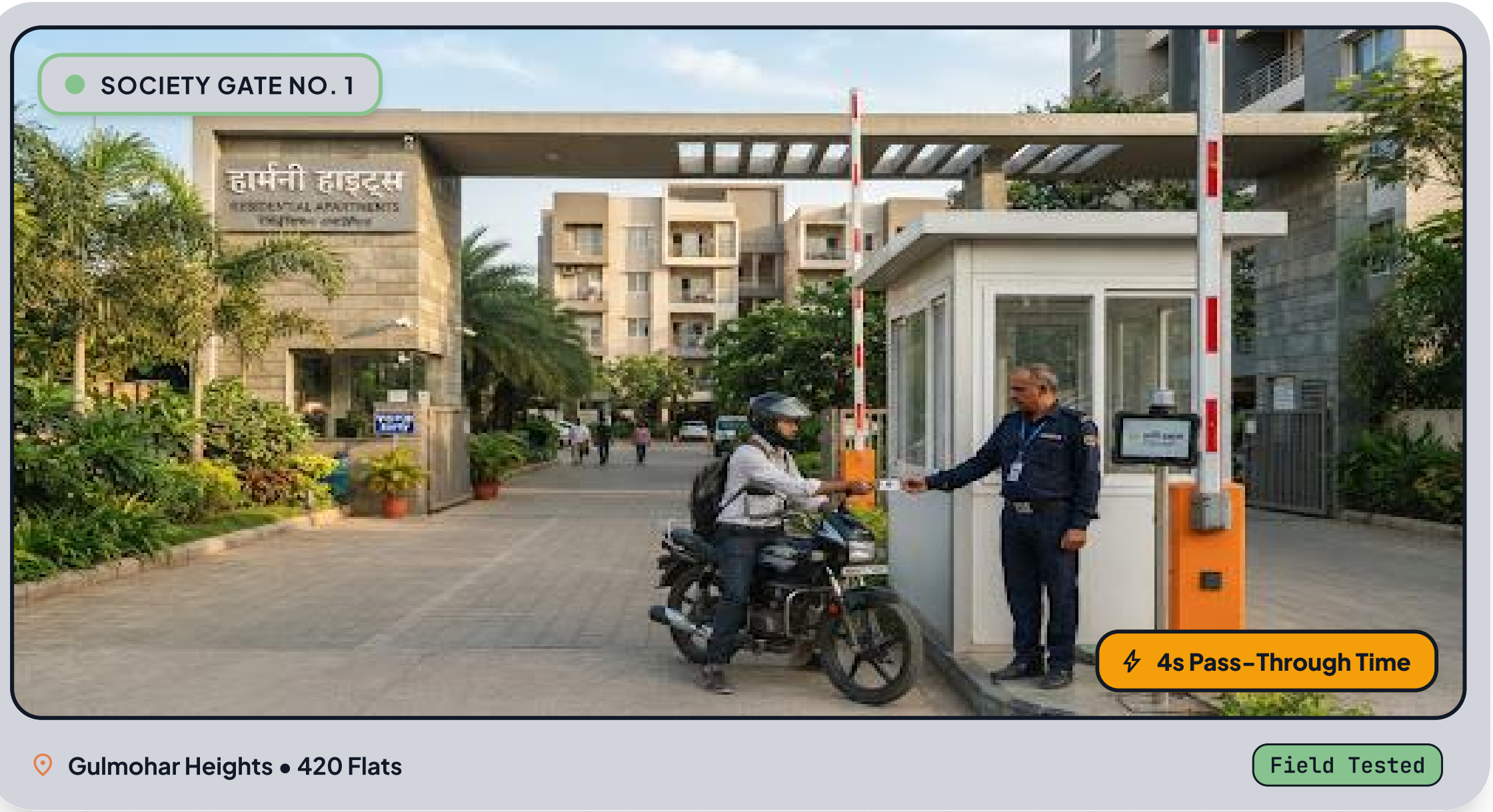
RESIDENTIAL VISITOR MANAGEMENT | Gate Clearance: <10 Seconds

The Gate-House Experience

Zero-friction visitor verification for Indian residential housing societies.

TIMELINE: 3 WEEKS

SKILLS INVOLVED:
- User Research
- Visual Design
- Interaction Design
- Rapid Prototyping



PRODUCT
Gatehouse App

METHODOLOGY
Contextual Inquiry & Rapid UX Prototyping

Problem, Need & Goal

Bridging the friction between high-security residential gates and rapid, dignified entry.

01 THE PROBLEM

GUARD FRICTION
Peak morning rush (8:45 AM) causes delivery bottlenecks, unreadable paper logbooks, and 45-60s delays per visitor.

VISITOR & RESIDENT TENSION
Awkward waiting in heat, unanswered intercom calls, and privacy violations from sharing personal phone numbers in public registers.

Physical logs average 38% invalid contact numbers (e.g. `9999999999`).

Avg. Verification Time 55s / visitor

02 THE USER NEED

GUARD (RAMESHWAR)
Glare-proof, zero-typing binary actions (Allow / Hold), 56px+ touch pads, bilingual Hindi/English guidance.

VISITOR (PRIYA)
Pre-approved QR fast-entry, no manual paper logging, transparent clearance status sent directly to mobile.

RESIDENT (HOMEOWNER)
Instant 1-tap push notifications without voice call disruptions during work meetings.

Core Workflow Expectation <10s gate clearance

03 APPLICATION GOAL

SPEED REDUCTION TARGET
Compress visitor gate clearance latency from 55 seconds down to under 10 seconds per verification.

AUDIT & DATA INTEGRITY
100% digital traceable audit trail with verified resident pre-approvals, replacing insecure physical paper.

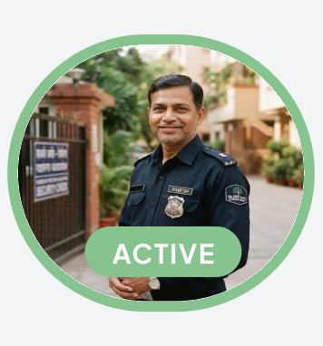
GUARD ERGONOMICS
Zero-cognitive load UI usable single-handed under high outdoor glare, requiring less than 2 minutes of staff training.

Success Benchmark <10s pass | 95%+ QR

● STAKEHOLDERS: Guards | Residents | Delivery & Visitors

Target Personas & Mental Models

Contrasting high-frequency gate operations against occasional guest expectations at high-density residential barriers.



Rameshwar Yadav (44)

LEAD GUARD • GULMOHAR HEIGHTS (420 FLATS)

8 yrs at society • 12-hr shifts • Redmi 9A user

PRIMARY GATE GOAL
"Clear the gate in under 15 seconds without unauthorized entry."

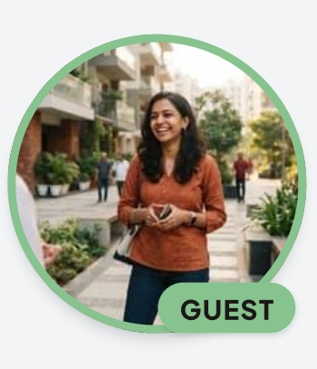
> "भैया, सुबह ९ बजे गेट पर आओ तो पता चले। एक हाथ में बैरियर की रस्सी, दूसरे में वॉकी-टॉकी, और पीछे चार गाड़ी हॉर्न बजा रही होती हैं। मुझे फैंसी बटन नहीं चाहिए बस बड़ा फोटो, फ्लैट नंबर, और एक हरा 'ALLOW' बटन।"
>
> *(Need 1-hand speed: just a clear photo, flat number, and a big green 'ALLOW' button.)*

✍ Fake Logbook Entries
Scribbled numbers (9999999999) leave guards blamed when parcels vanish.

📞 Intercom Bottleneck
Unanswered flat calls cause delivery bikes to pile up while cars honk in line.

KEY UX DECISIONS

≥56px	2-Tap	Bilingual
Thumb Targets	Photo + Flat Entry	Hindi & Color Cues



Priya Sharma (25)

ONE-TIME SOCIAL VISITOR • WEEKEND GUEST

Visiting friend at Gulmohar Heights (Flat 702)

Frequency: First-time visit (Rare / Occasional)

📱 Smartphone Savvy — High Proficiency
Uses iPhone 14 & WhatsApp fluently. Strict zero-app tolerance: will NOT download an app for a single social visit.

> "यार, मैं सिर्फ शनिवार की शाम को कॉफी पीने आई हूँ। मुझे गार्ड को अपना फोन नंबर रजिस्टर में नहीं लिखवाना, न ही 10 मिनट धूप में गेट पर खड़े रहना है। बस एक WhatsApp लिंक या QR कोड हो, दिखाओ और अंदर जाओ।"
>
> *Need : Send me a WhatsApp QR pass, beep, and let me in.")*

Privacy Concern: Public Logbook
Forced to write personal phone number & name in a register where anyone behind her can see and copy it.
Risk of unsolicited spam / calls

⌛ The Awkward Intercom Waiting Trap
Stranded outside the gate in a cab while the guard repeatedly calls an unanswered flat intercom.
Awkward social friction at gate

PRIMARY VISITOR GOAL
"Zero-friction entry in under 10 seconds — without awkward interrogations or handing over private data."

Target: <10s Gate Pass

Existing User Journeys & The "Pre-Approval" Illusion

Deconstructing why current gate clearance fails: from high-friction guard phone calling to the excessive cognitive chore of manual resident QR pass generation.

A Traditional Intercom / Phone Flow

Reactive, Latent & Single-Point Vulnerability

Latency: 60s - 90s | Failure Rate: 35%

Guard Rameshwar Yadav GATE ATTENDANT
"Every evening 40 delivery bikes queue up. Residents don't answer their phones, drivers shout, and I have to scrawl illegible numbers in torn logbooks."

1. **Visitor Arrival & Intercom Dial**
   Visitor halts at boom barrier, shouts flat number. Guard flips intercom log and dials the resident unit.
2. **Unanswered Ring & Queue Bottleneck** High Friction
   Resident is in a meeting or phone is on silent. 2 cars & 3 delivery bikes stack up behind the barrier, honking persistently.
3. **Awkward Interrogation & Paper Register**
   Guard dials personal phone. Visitor hurriedly scrawls unverified fake phone numbers on torn, unsearchable paper logbook.
4. **Delayed Barrier Lift (70+ Seconds)**
   Resident finally answers: "Who is it again?". Guard presses manual button; impatience causes tailgating & security breaches.

⚠ **Core Vulnerability:** Extreme guard anxiety during peak 7-9 PM rushes, loud honking gates, and zero visitor credential verification.

B Current Manual QR Code Flow

The Resident Burden: "Fast at Gate, Agonizing in Prep"

Gate Speed: 8s | Adoption Drop: 70%

Priya Sharma APARTMENT RESIDENT
"I just ordered groceries on Blinkit. Why would I spend 3 minutes filling 6 fields in a community app to create a guest pass when the delivery takes 8 minutes total?"

1. **Unrealistic Pre-Planning Demand**
   Requires resident to predict visits in advance. Completely unrealistic for rapid, spontaneous Swiggy, Blinkit, or Amazon deliveries.
2. **6-Screen App Labyrinth** High Effort
   App open → "Gate Passes" → "Add Visitor" → Select Category → Fill 6 text inputs → Date/Time slot slider.
3. **Manual WhatsApp Relay Export**
   App generates image badge. Resident switches app, searches visitor on WhatsApp, and forwards static image pass.
4. **Instant 2s Scan (The ONLY Fast Touchpoint)** 2s Scan
   Guard scans screen with tablet. Boom barrier opens instantly. But 88% of visits never reached this stage.

The Illusion: While the gate scan itself takes only ~8s, residents spend nearly 3 minutes of friction to set it up beforehand!

CRITICAL FRICTION GAP ⚡ PRE-APPROVAL FATIGUE

Why do 88% of residents still force guards to call them?

Because opening an app, typing visitor full names, selecting vehicle category, and sharing QR passes requires 18 distinct micro-taps for an impulse ₹150 delivery. The cognitive cost completely outweighs the gate clearance benefit.

6+	2.8m	12%
REQUIRED FIELDS	AVERAGE SETUP	DELIVERIES PRE-DONE
Per manual pass generation	Time before visitor arrival	88% drop into manual call

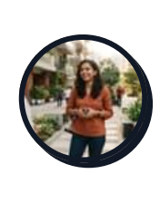
UX OPPORTUNITY: Shift from manual upfront creation to 1-tap auto-sync from SMS / e-commerce notifications. TARGET: <3 SEC

Bridging the Gap: The Zero-Friction AI Experience Layer

The paradigm shift from active manual paperwork to ambient intelligence: eliminating high-friction pre-approval forms via instant voice synthesis and background webhook sync.

MANUAL FRICTION ~~2.8 mins~~ → AMBIENT AI LAYER < 1 sec

● THE COGNITIVE GAP — `88% Drop-Off Rate`



Priya S. Resident & High-Frequency Requester
"I'm ordering lunch while in a client meeting. Having to type out a delivery person's vehicle number and choose a 30-minute window for a ₹180 sandwich is absurd—I'd rather just let the guard ring."

01 **6 Mandatory Input Fields**
Name, phone, vehicle type, license plate, gate designation, and arrival expiry window create high cognitive fatigue.

02 **Destructive Context Switching**
Residents must navigate out of Blinkit/Swiggy or chat apps into Gatehouse, manually copy OTPs/names, then toggle back.

03 **Dynamic Traffic & Expired Passes**
City traffic spikes invalidate rigid 15-minute scheduled windows, prompting guard intercoms anyway.

● DUAL-MODALITY AI LAYER — Voice Native • Ambient Sync

Conversational Voice Pass
For Friends & Social Guests
Speak intent into Siri, Assistant, or in-app mic. Passes are derived and dispatched instantly.
"Pass for Rahul at 8 PM tonight"

Ambient Detection
For Deliveries & Couriers
Detects dispatch webhooks & order SMS from Blinkit, Zepto, and Swiggy with zero app interaction.
SMS & Webhook Parsers Active `0-TOUCH`

INTELLIGENT PIPELINE

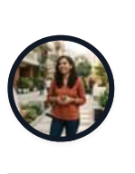
`PHASE 01` **Ingest Stream**
Captures voice audio or background SMS telemetry without app launch.

`PHASE 02` **Entity Parsing**
On-device LLM extracts guest identity, time window, and target flat number.

`PHASE 03` **Silent Handshake**
Ramesh's tablet auto-validates QR pass. Zero resident calls triggered.

User Journey Flows: Frictionless Clearance Ecosystem

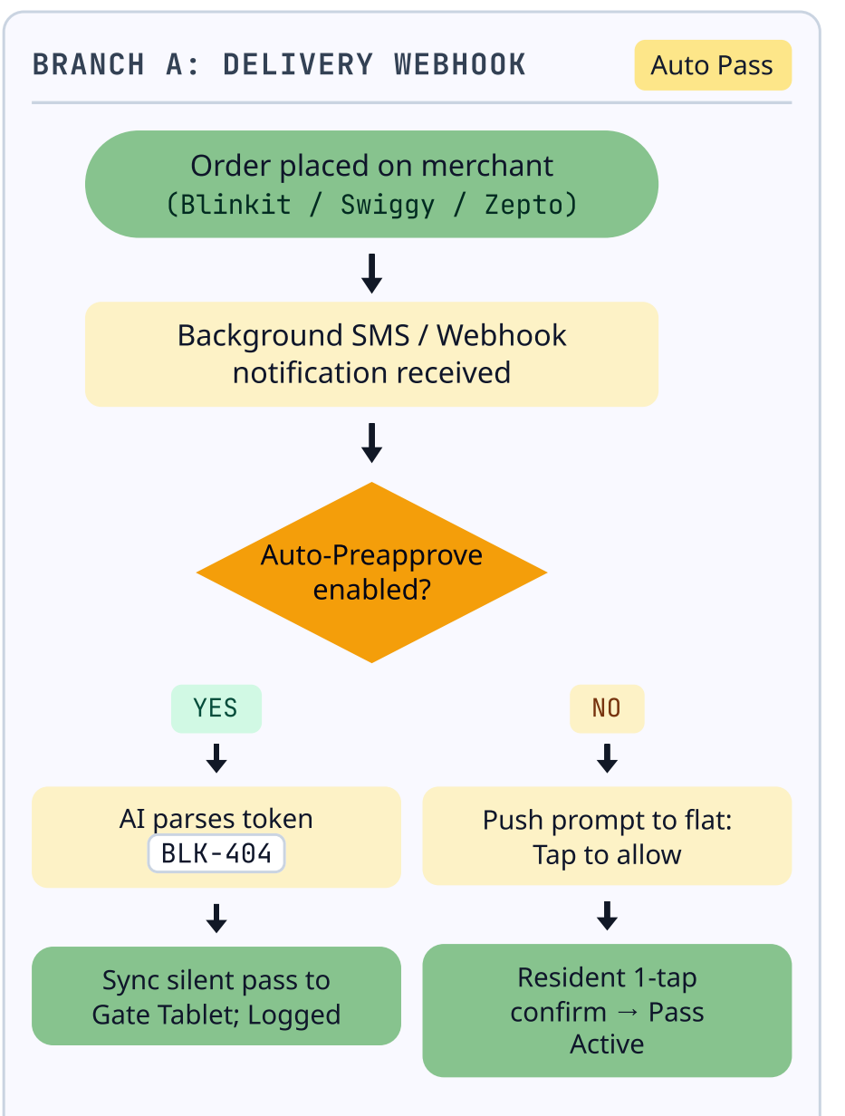
Mapping the synchronous interaction loops between the Resident app, Ambient AI parsing, and Guard tablet clearance across delivery and guest scenarios.

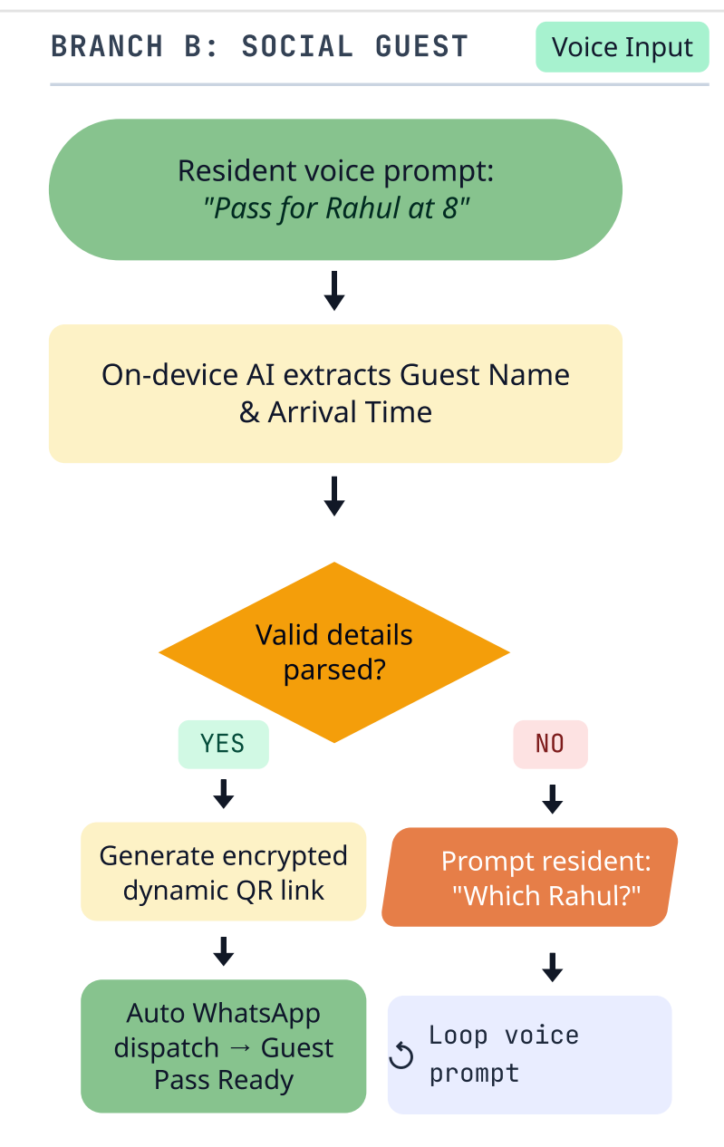


1. Resident Flow Topology

`Priya Sharma` — 🎙 Ambient & Voice

Flat 404 • Resident Host Persona & Voice Assistant

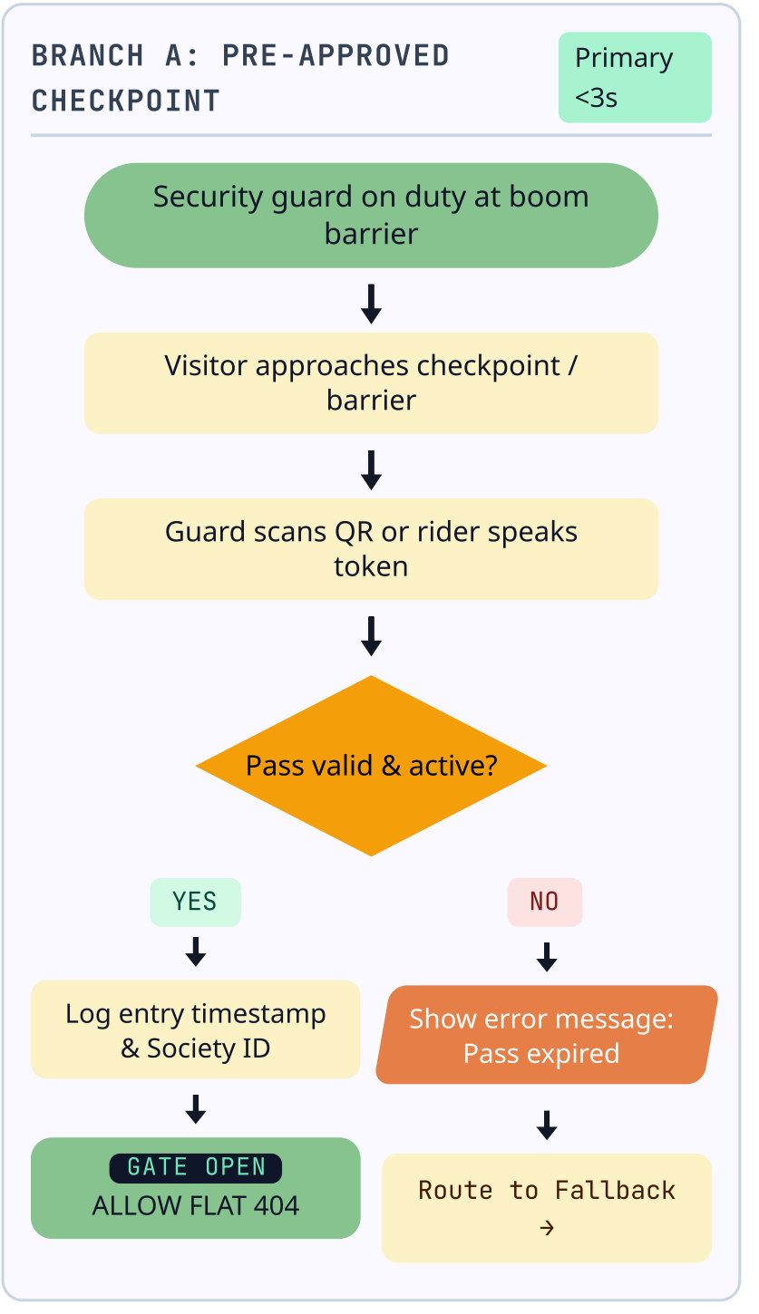

BRANCH A: DELIVERY WEBHOOK
Auto Pass
Order placed on merchant (Blinkit / Swiggy / Zepto)
Background SMS / Webhook notification received
Auto-Preapprove enabled?
YES
NO
AI parses token BLK-404
Push prompt to flat: Tap to allow
Sync silent pass to Gate Tablet; Logged
Resident 1-tap confirm → Pass Active

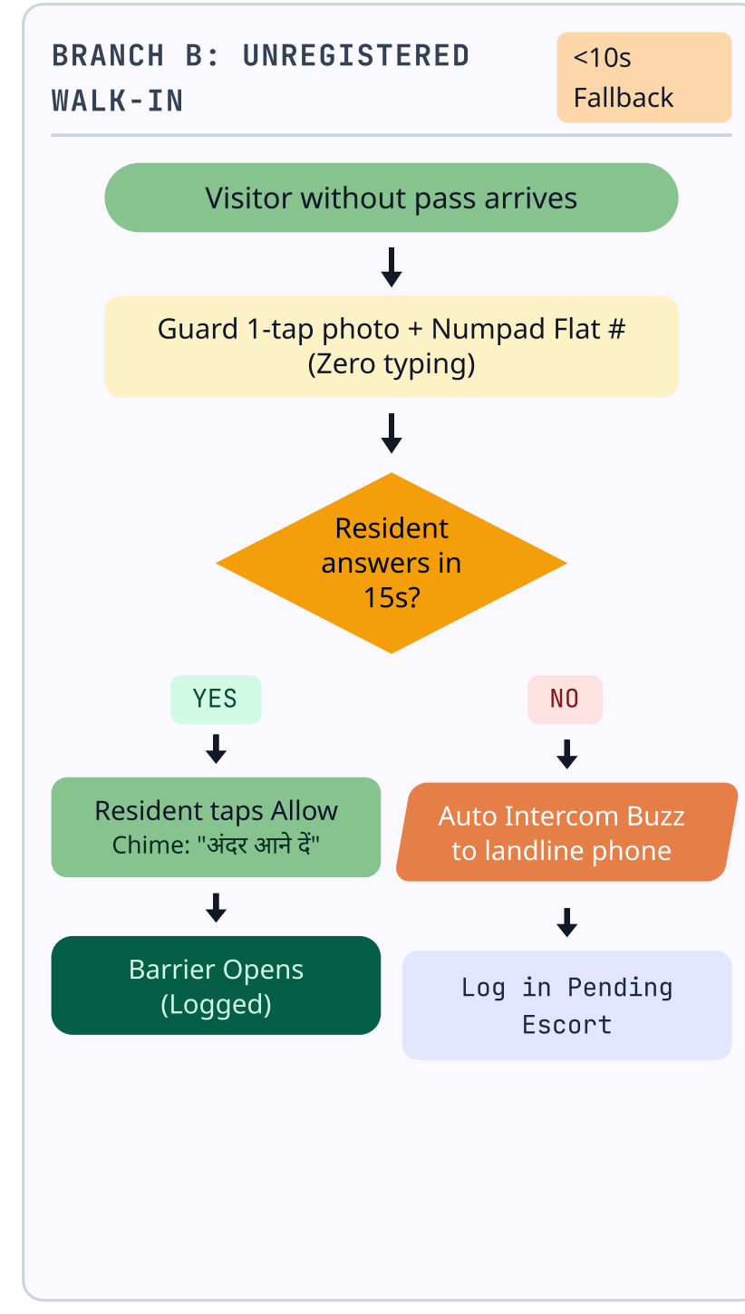

BRANCH B: SOCIAL GUEST
Voice Input
Resident voice prompt: "Pass for Rahul at 8"
On-device AI extracts Guest Name & Arrival Time
Valid details parsed?
YES
NO
Generate encrypted dynamic QR link
Prompt resident: "Which Rahul?"
Auto WhatsApp dispatch → Guest Pass Ready
Loop voice prompt

2. Security Guard Clearance Logic

`Rameshwar Yadav`

Main Gate Post


BRANCH A: PRE-APPROVED CHECKPOINT
Primary <3s
Security guard on duty at boom barrier
Visitor approaches checkpoint / barrier
Guard scans QR or rider speaks token
Pass valid & active?
YES
NO
Log entry timestamp & Society ID
Show error message: Pass expired
GATE OPEN
ALLOW FLAT 404
Route to Fallback →


BRANCH B: UNREGISTERED WALK-IN
<10s Fallback
Visitor without pass arrives
Guard 1-tap photo + Numpad Flat # (Zero typing)
Resident answers in 15s?
YES
NO
Resident taps Allow Chime: "अंदर आने दें"
Auto Intercom Buzz to landline phone
Barrier Opens (Logged)
Log in Pending Escort